Learning Risk-Aware Policies: A Comparative Study of Reinforcement Learning and Heuristic Agents in Flip 7

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***Abstract*—This study investigates whether a tabular Q-learning agent can discover risk-aware, position-sensitive stopping policies in a stochastic multiplayer card game without explicit programming of decision thresholds. Flip 7 is modeled as a Markov Decision Process in which the agent learns to select between drawing and stopping against three calibrated heuristic opponents representing Conservative, Greedy, and Balanced risk profiles. The agent employs a four-dimensional state representation encoding round score bins, cards drawn, competitive score gap, and second chance availability, combined with a hybrid reward function that balances normalized score accumulation, positional awareness, and tiered bust penalties. Trained over 10,000 simulated games and evaluated across 10 independent folds, the agent achieved a mean win rate of 0.167 ± 0.028, exceeding the Conservative heuristic but falling significantly below the Greedy baseline ($t = -11.41$, $p < 0.0001$). Results demonstrate that reward function asymmetry and tabular state abstraction represent compounding limitations that prevent the agent from learning sufficiently aggressive scoring behavior. These findings motivate future transitions to Deep Q-Network architectures capable of encoding richer game state information without dimensionality constraints.**

I. Introduction

Optimal stopping problems appear in finance, operations research, and game theory, where agents must determine when continued action is no longer worth the risk. As reinforcement learning methods mature, a natural question emerges: can an agent learn adaptive stopping policies in stochastic, competitive environments without being explicitly programmed with decision thresholds? In multi-player stochastic environments, static heuristics may fail to account for scenarios where the objective can shift from maximizing an expected score when behind to maximizing the probability of victory when ahead. Unlike classical optimal stopping models that assume a single decision maker and a fixed objective, competitive card games introduce dynamic positional states where the optimal action depends not only on the agent's current score but also on its standing relative to opponents. Flip 7 is modeled as a Markov Decision Process (MDP) where the learning agent interacts with a stochastic environment against three heuristic opponents. The agent is trained using tabular Q-learning with a state representation encoding round score bins, cards drawn count, score gap relative to the leading opponent, and second chance card availability. Opponents follow fixed heuristic policies representing Conservative, Balanced, and Greedy difficulty configurations. Within this environment, the agent learns to select between two actions, DRAW or STOP, over 10,000 simulated games. The agent demonstrated position-sensitive behavior by drawing more aggressively when significantly behind opponents. The agent obtained a win rate of 16.2%, higher than the Conservative heuristic of 15.3%, however, it greatly underperformed the Greedy heuristic baseline at 25.1%, a gap attributed primarily to the reward function asymmetry. Immediate bust penalties outweighed sparse win-based rewards, driving the agent toward a conservative stopping behavior that limited average points banked per round to 18.5, which is inefficient to competitively reach the 200 point win threshold against the provided heuristic agents. This paper makes the following contributions. First, we construct a decision-aware state representation using score bins aligned to empirically observed heuristic decision boundaries, replacing raw score tracking to improve Q-table state coverage and ensure the agent visits each state enough times to build reliable learned values. Second, we design a hybrid reward function that combines normalized score accumulation with positional awareness, rewarding the agent more for banking points when trailing and applying steeper penalties for reckless play when already in a winning position. Third, we identify two compounding limitations that prevented the agent from reaching competitive win rates. The reward function punished busting immediately and concretely, but winning the game only produced a single reward signal at the very end, meaning the agent learned that stopping safely was better than risking a bust even when drawing more cards was the right strategic choice. Additionally, tabular Q-learning cannot distinguish between two situations that look identical on paper but carry very different risks, such as having already drawn high value cards that are likely to appear again versus holding low value cards with little bust exposure. Architectures such as Deep Q-Networks, which use neural networks to process richer game information, represent a natural path forward for addressing this limitation.

II. Background

A. Russell and Norvig (2021)

Russell and Norvig (2021) establish Markov Decision Processes (MDPs) as the formal framework for sequential decision-making under uncertainty, defining the core components of states, actions, transitions, and rewards. This framework directly structures the Flip 7 agent, where game situations map to discrete states, the agent selects between drawing and stopping at each turn, the card deck provides stochastic transitions, and a hybrid reward function evaluates outcomes. The MDP formulation provides the theoretical foundation upon which the Q-learning algorithm operates, connecting classical AI planning theory to the practical implementation developed in this project.

B. Watkins and Dayan (1992)

Watkins and Dayan (1992) establish Q-learning as a model-free reinforcement learning method with a mathematical proof guaranteeing convergence to an optimal policy in any finite MDP. To understand Q-learning, it helps to first distinguish between on-policy and off-policy learning. In standard on-policy reinforcement learning, an agent learns the value of the exact policy it is currently following, meaning its estimates are tied to its own behavior. Q-learning is off-policy, meaning the agent learns the value of the optimal policy independently of how it currently behaves during exploration. This distinction is critical in stochastic environments like Flip 7, where an agent must explore suboptimal actions early in training to eventually discover which decisions lead to winning outcomes. Their Bellman update rule allows the agent to refine action-value estimates by incorporating observed rewards and discounted future expectations after every decision. Two parameters govern the stability and efficiency of this learning process: the learning rate, which controls how aggressively the agent updates its estimates from new experience, and the discount factor, which determines how much the agent weighs future rewards relative to immediate ones. Watkins and Dayan demonstrate that with appropriate parameter selection, Q-learning is guaranteed to converge given sufficient state visitation, a requirement directly addressed in this project by constraining the agent to a finite set of discrete state-action pairs.

C. Barros, Tanevska, and Sciutti (2020)

Barros, Tanevska, and Sciutti (2020) propose a reinforcement learning framework for competitive card games where agents must remain ”socially aware” of opponent behavior rather than optimizing purely for individual score maximization. Their work directly informs the positional awareness component of this project, where the agent evaluates the total point difference between itself and the leading opponent to dynamically adjust its risk tolerance based on competitive standing. Rather than treating each round in isolation, both their framework and this project recognize that optimal stopping behavior shifts depending on whether the agent is leading or trailing. They also identify the ”exploration-exploitation” tradeoff as a core challenge, noting that excessive exploration can lead to costly failures. In Flip 7, this displays directly as bust events, where an agent that draws too aggressively loses all points accumulated that round. This project addresses the tradeoff through epsilon-greedy exploration with a decaying exploration rate, gradually shifting the agent from broad exploration early in training toward exploiting its learned policy as training progresses.

D. Jang et al. (2019)

Jang et al. (2019) provide a comprehensive classification of Q-learning variants, spanning from standard tabular implementations to Deep Q-Networks and other advanced architectures. Their framework distinguishes value-based methods, which learn explicit action-value estimates, from policy-based and actor-critic approaches, providing a principled basis for algorithm selection. This project adopts tabular Q-learning specifically for its transparency — every learned value can be inspected directly in the Q-table, making it possible to analyze exactly what stopping behavior the agent discovered and why. Jang et al. also highlight that tabular methods require careful state abstraction in complex environments to avoid the curse of dimensionality, where an explosion of unique states prevents the agent from visiting any single state enough times to build reliable estimates. This tradeoff is directly confronted in this project through decision-aware score binning and gap binning, which reduce the state space to 513 well-visited states at the cost of card-level information the agent cannot reason about. As their review suggests, transitioning to Deep Q-Networks would allow richer state representations without the dimensionality constraint, a direction identified as the primary avenue for future work.

E. Sturtevant (2003)

Sturtevant (2003) bridges classical single-agent reinforcement learning, which typically optimizes against a static environment, and the more complex dynamics that emerge in multiplayer competitive settings. Sturtevants work demonstrates that optimal behavior in multiplayer games is not static, as an agent that maximizes its own expected score may still lose if it fails to account for the relative standings of opponents. This information directly motivates the positional awareness component of this project, where the state representation encodes the score gap between the agent and the leading opponent rather than tracking only the agent’s own round score. By incorporating competitive standing into the learning pool, the agent moves beyond self-centered score maximization toward the kind of adaptive, opponent-aware decision making that Sturtevant identifies as essential in multiplayer algorithmic approaches.

III. METHODOLOGY

Flip 7 is a card game that balances strategy with risk. The objective is to be the first player to reach 200 or more points by drawing cards from a randomized deck. Points are scored by the face value of each card drawn. The numbered cards follow a distribution where each value appears a number of times equal to its face value, with one card valued at 1, two cards valued at 2, continuing through twelve cards valued at 12, plus a single zero card, for a total of 95 numbered and special cards. The deck also includes special cards that can help or hinder any player. The "+x" bonus cards add a fixed point value to the drawing player's current round score, while the "x2" card doubles the player's total round score at the time it is drawn. The Freeze card stops a targeted opponent from drawing further that round, locking in their current score. The Draw 3 card forces a targeted player to immediately draw three consecutive cards. The Second Chance card grants the drawing player immunity from one bust event during that round. The bust mechanic is central to the game's risk structure. If a player draws a number they have already drawn in the same round, they immediately bust, losing all points accumulated that round and sitting out until the round concludes.

Outside of special card effects, each player faces exactly two choices on their turn: draw another card and risk busting, or stop and bank their current round points toward their cumulative total. This binary decision under uncertainty is the core optimal stopping problem this project investigates. This implementation adopts several simplifications relative to the published Flip 7 rules to maintain focus on the core stopping decision. The deck configuration uses two Second Chance cards rather than three, two each of the +2 and +4 bonus cards rather than one, and three each of the Freeze and Draw 3 cards. The 15-point bonus awarded for drawing exactly 7 cards in a round is excluded to avoid incentivizing a fixed card count target rather than adaptive stopping behavior. Additionally, the win condition is implemented as immediate victory upon any player reaching 200 total points, rather than completing the current round and awarding victory to the highest scorer. These simplifications reduce stochastic complexity while preserving the fundamental risk-reward tradeoff that motivates this study. Alignment with official game rules represents a natural direction for future work. For this project, Flip 7 is modeled as a Markov Decision Process where the learning agent interacts with a stochastic environment against three heuristic opponents. The state space, action space, transition dynamics, and reward function are each described in detail in the following subsections. The complete system architecture is shown in Fig. 1.

A. Heuristic Agents

Three heuristic policies were designed to serve as controlled baseline opponents, each representing a distinct risk tolerance profile. The Conservative policy minimizes bust risk by stopping when the round score reaches 18 points or when 4 cards have been drawn, whichever occurs first. This threshold was selected to mirror cautious play that prioritizes banking small but reliable point totals each round. In 3-player evaluations, the conservative win rate was a strong 34.2%. That win rate fell in 4-player baseline evaluation over 5,000 games as the Conservative agent achieved a win rate of 15.3% and averaged 21.6 points banked per stop. The Greedy policy pursues aggressive score accumulation by continuing to draw until its round score reaches 30 points or 6 cards have been drawn. This configuration intentionally accepts a higher bust risk in exchange for larger banking opportunities. In a 4-player baseline evaluation the Greedy agent achieved a win rate of 25.1% and averaged 33.3 points banked per stop, making it the strongest performing heuristic and the primary comparison target for statistical evaluation. The Balanced policy occupies the middle ground between the two extremes, stopping at a round score of 26 points or 6 cards drawn. Initial baseline testing revealed that the original threshold of 23 points produced a win rate of 26.6% in 3-player evaluation, underperforming relative to its risk profile. The threshold was empirically tuned to 26 points through iterative baseline analysis, which improved average points banked per stop from 26.7 to 29.9 and increased the 4-player win rate to 23.5%. This tuning process demonstrated that small threshold adjustments produce measurable behavioral differences, a finding that contextualizes the sensitivity of the RL agent's learned policy to its reward signal design.

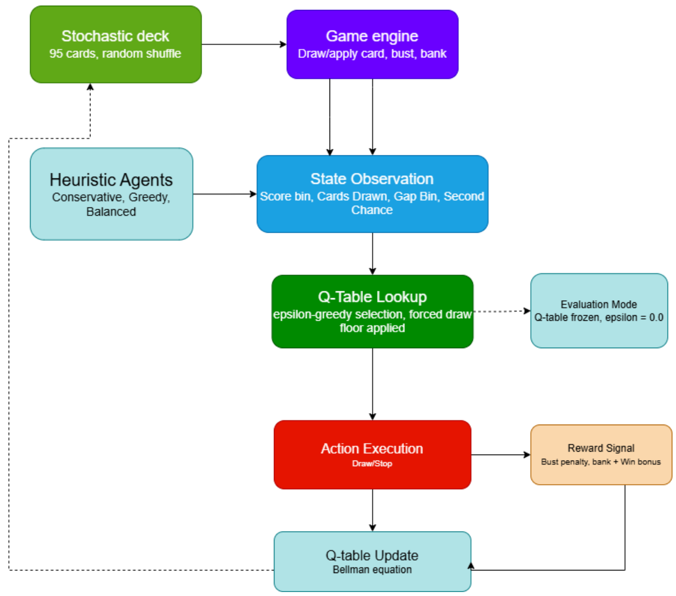

Fig. 1. Flip 7 RL system architecture — training loop.

B. State Representation

The state space is represented as a four-dimensional tuple $(s_{bin}, c_{drawn}, g_{bin}, sc)$ where each dimension captures a distinct aspect of the agent's decision context at each turn. The first dimension, s_{bin}, encodes the agent's current round score as one of eight discrete bins rather than a raw integer value. Initial implementation tracked exact round scores, producing over 2,500 unique states across training, which is too sparse for reliable Q-value estimation given a 10,000 game training budget. Replacing raw scores with decision-aware bins reduced the state space to 513 well-visited states, increasing average state visitation from approximately 59 visits to over 290 visits per state. The bin boundaries were deliberately aligned with empirically observed heuristic stopping thresholds established during baseline evaluation:

TABLE I
SCORE BIN DEFINITIONS

Bin	Range	Strategic Zone
0	0–12	Early draw zone
1	13–18	Conservative threshold
2	19–23	Balanced threshold
3	24–29	Greedy threshold
4	30–36	High risk zone
5	37–42	Very high risk
6	43–49	Extreme risk
7	50+	Near-certain stop

The second dimension,c_{drawn}, tracks the number of cards drawn in the current round. As more cards are drawn from a finite deck, the probability of drawing a duplicate number increases, making bust risk a function of draw count as well as score. This dimension allows the agent to incorporate implicit bust risk into its stopping decision without explicit deck tracking.

The third dimension, g_{bin}, encodes the agent's positional awareness relative to the leading opponent. Motivated by Sturtevant (2003) and Barros et al. (2020), this dimension allows the agent to shift its risk tolerance based on competitive standing. Four gap bins are defined in Table II below.

TABLE II
GAP BIN DEFINITIONS

Bin	Condition	Interpretation
0	Agent leads by >15 pts	Ahead
1	Gap within ± 15 pts	Roughly even
2	Trailing by 16–45 pts	Moderately behind
3	Trailing by >45 pts	Significantly behind

The fourth dimension, sc, is a binary indicator of Second Chance card availability. The Second Chance card nullifies one bust event, effectively reducing the risk of drawing an additional card. Including this dimension allows the agent to learn more aggressive drawing behavior when a safety net is available, and more conservative behavior when it is not.

C. Reward Function

The reward function combines three signals to shape the agent's stopping behavior: a normalized score component, a positional awareness component, and a tiered bust penalty.

When the agent successfully banks points, the reward is computed as:

$$r = 0.6 \cdot \frac{p_{banked}}{200} + 0.4 \cdot \left(-\frac{g_{raw}}{200}\right) \tag{1}$$

where p_{banked} is the round score banked and g_{raw} is the raw score gap between the leading opponent and the agent. A positive gap indicates the agent is trailing, reducing the reward to create pressure for more aggressive play. A negative gap indicates the agent is leading, reinforcing banking behavior when ahead. The reward is capped at 1.0 to maintain bounded Q-values.

Bust penalties are applied as flat negative rewards with tiered severity based on competitive position at time of bust:

$$r_{bust} = \begin{cases} -0.6 & \text{if ahead by } \geq 45 \text{ pts (greed tax)} \\ -0.45 & \text{if ahead by 20–44 pts} \\ -0.3 & \text{otherwise (trailing or even)} \end{cases} \tag{2}$$

The tiered structure avoids penalizing aggressive play when trailing, recognizing that high-risk draws represent rational comeback attempts. A fixed win bonus of $r_{win} = +5.0$ is applied when the agent wins the game, delivered as a terminal Q-table update to the final state-action pair.

D. Q-Learning Agent

The Q-table stores estimated action values for every state-action pair the agent encounters, mapping each four-dimensional state tuple to separate value estimates for DRAW and STOP. All values are initialized to zero, ensuring the agent enters training without prior assumptions about which actions are preferable in any situation. An epsilon-greedy strategy governs action selection, encouraging broad exploration early in training when the Q-table contains little reliable information, and gradually shifting toward exploitation of learned values as training progresses. Actions are binary as the agent chooses to DRAW or STOP at each turn. A forced draw floor was introduced to prevent early stopping, requiring the agent to draw at least twice and accumulate a score above bin zero before STOP becomes an eligible action.

Q-values are updated after every action using the Bellman equation:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right] \tag{3}$$

The learning rate $\alpha = 0.7$ controls how aggressively the agent incorporates new experience into its existing estimates, a high value deliberately chosen to accelerate learning across a fixed 10,000 game training budget. The discount factor $\gamma = 0.63$ governs the relative weight the agent places on future rewards versus immediate ones, producing a slightly future-leaning policy that still values present outcomes. When the agent survives a draw, the update incorporates both the immediate reward and the discounted value of the best available action in the resulting state. When the agent busts, the round ends and no future state exists, as the future reward term is dropped entirely and the agent learns directly from the bust penalty alone. For terminal transitions this reduces to:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r - Q(s,a) \right] \tag{4}$$

E. Training Procedure

The agent is trained over 10,000 simulated games against three heuristic opponents in a 4-player configuration. The exploration rate decays after each game according to:

$$\epsilon_{t+1} = \max(\epsilon_{min}, \epsilon_t \cdot \delta) \tag{5}$$

where $\delta = 0.9996$ and $\epsilon_{min} = 0.05$. The agent reaches the exploration floor at approximately game 8,000, after which it continues to explore at a fixed 5% rate rather than stopping exploration entirely. Turn order is randomized each cycle within a round to eliminate first-turn positional advantage. Special card effects including Freeze and Draw 3 are resolved using fixed heuristic rules to limit state explosion while preserving core game dynamics.

Upon completion of training the Q-table is frozen and the agent shifts to evaluation mode, so that all decisions are made purely by exploiting learned Q-values with no random exploration. The trained Q-table uses Python's pickle library, allowing evaluation to be completed without retraining.

IV. Experiments and Rigorous Evaluation

A. Evaluation Metrics

Four metrics were selected to evaluate agent performance from both a competitive and behavioral perspective. Win rate performs as the primary metric, measuring the agent's ability to outperform heuristic opponents and reach the 200 point threshold first. Bust rate per round captures how frequently the agent loses accumulated points through overly aggressive drawing, providing insight into whether the reward function successfully shaped risk-aware behavior. Average points banked per stop measures scoring efficiency, revealing whether the agent learned to stop at meaningful score levels rather than banking small amounts. Finally, Q-table state coverage tracks the number of unique states visited during training, confirming that the state representation design allowed sufficient exploration of the decision space without sparse visitation. Together, these four metrics distinguish between an agent that wins by luck versus one that has learned adaptive stopping behavior.

B. Experimental Setup & Hypothesis

The null hypothesis posits that the Q-learning agent's win rate is statistically equivalent to that of the Greedy heuristic baseline:

$$H_0 : \mu_{RL} = \mu_{Greedy} \tag{6}$$

$$H_1 : \mu_{RL} \neq \mu_{Greedy} \tag{7}$$

A 10-fold paired t-test was conducted to evaluate this hypothesis, retraining the agent from scratch on each fold using a unique random seed to ensure results reflect consistent learned behavior rather than a single favorable initialization. Each fold evaluated 200 games per agent, yielding 2,000 total evaluation games across all folds. The Greedy heuristic was selected as the primary comparison target as it achieved the highest win rate among all baseline agents in 4-player evaluation.

C. Heuristic Baseline

Initial 3-player baseline evaluation produced competitive win rates across all three heuristics, confirming that the stopping thresholds were appropriately calibrated. In 4-player evaluation over 5,000 games, with the RL agent seat filled by a second Balanced agent to establish a fair comparison benchmark, the Greedy heuristic achieved the highest win rate at 26.2%, followed by Balanced at 23.1% and Conservative at 14.9%. An unexpected performance asymmetry was observed in the Balanced_2 agent occupying seat zero, which achieved a 36.9% win rate despite running an identical policy to the Balanced agent. This anomaly is hypothesized to result from special card targeting bias favoring players with lower accumulated scores and is documented as a known limitation. Detailed behavioral metrics are reported in Table III.

TABLE III
4-Player Heuristic Baseline Behavioral Metrics

Agent	Avg Banked	Bust Rate	Stop Rate
Conservative	21.7	0.046	0.224
Greedy	33.2	0.089	0.119
Balanced	29.9	0.075	0.145

1) RL Agent Results: After multiple iterative training runs addressing reward scaling and state representation design, the final RL agent achieved a win rate of 15.3% in primary evaluation over 2,000 games. As shown in Table IV, the agent underperformed all three heuristic opponents, with Greedy achieving the highest win rate at 32.7%, followed by Conservative at 26.6% and Balanced at 25.4%.

Fig. 2 compares bust rates across all agents. The agent's average points banked per stop of 18.4 and bust rate of 0.110 per round suggest the agent learned to stop conservatively at low score totals, prioritizing avoidance of immediate bust penalties over accumulating the larger round scores necessary to reach the 200 point win threshold competitively.

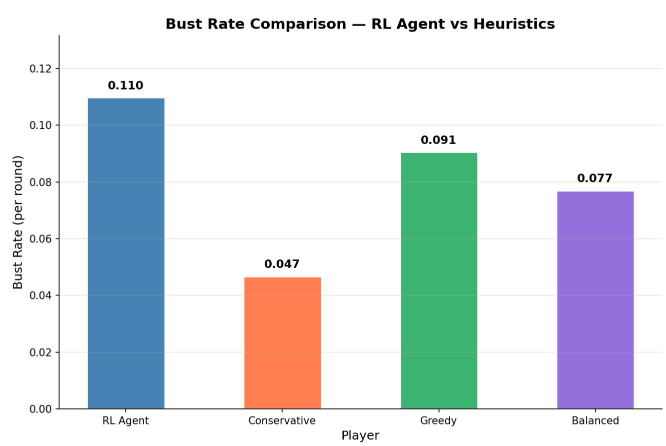

Fig. 2. Bust rate per round comparison across all agents.

Fig. 3 shows the RL agent win rate and Q-table state coverage across 10,000 training games.

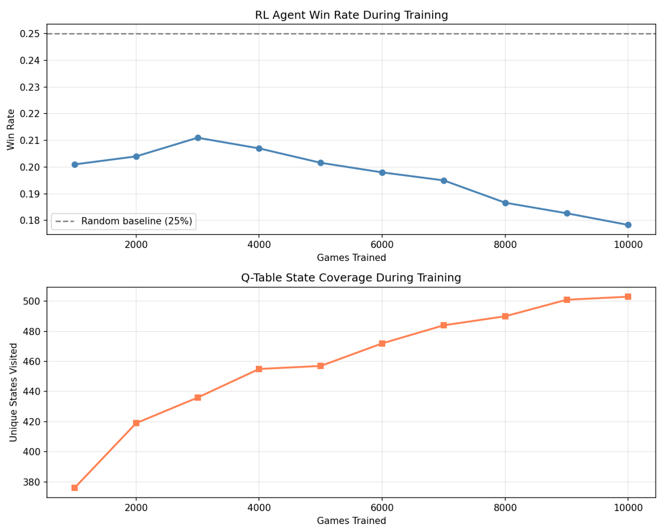

Fig. 3. RL Agent win rate and Q-table state coverage during training.

TABLE IV
RL AGENT VS HEURISTIC WIN RATES (2000 GAMES)

Agent	Win Rate	Avg Banked	Bust Rate
RL Agent	0.153	18.4	0.110
Conservative	0.266	21.6	0.047
Greedy	0.327	33.3	0.091
Balanced	0.254	29.9	0.077

2) Statistical Evaluation: A 10-fold paired t-test comparing the RL agent against the Greedy heuristic across independent training runs yielded a t-statistic of $t = -11.41$ ($p < 0.0001$), rejecting the null hypothesis and confirming a statistically significant difference in win rates. The mean RL agent win rate across folds was 0.167 ± 0.028 with a 95% confidence interval of $(0.146, 0.189)$, compared to a Greedy mean of 0.345. The negative direction of the t-statistic confirms that the Greedy heuristic consistently and significantly outperforms the RL agent across all training seeds. These results suggest that while the agent learned a stable policy, further refinement of the reward function and state representation are necessary to close the performance gap against aggressive heuristic opponents.

TABLE V
STATISTICAL TEST RESULTS (K=10 FOLDS)

Metric	Value
Mean RL Win Rate	0.167 ± 0.028
Mean Greedy Win Rate	0.345
95% Confidence Interval	(0.146, 0.189)
t-statistic	-11.4147
p-value (two-sided)	$p < 0.0001$
p-value (one-sided)	$p < 0.0001$
Significant ($p < 0.05$)	Yes

Fig. 4 illustrates the scoring efficiency gap between the RL agent and heuristic opponents. Fig. 5 visualizes learned action preferences across the state space.

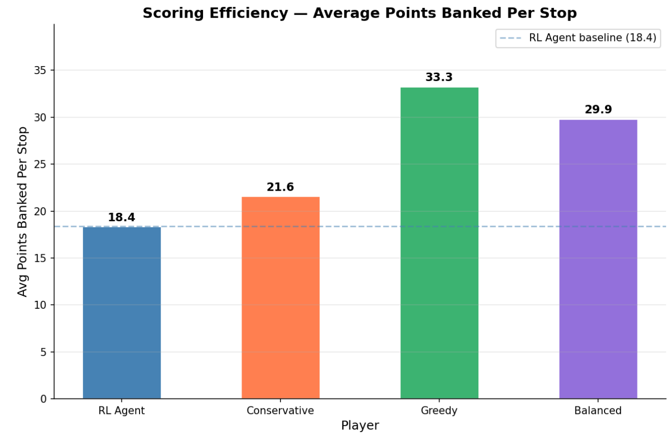

Fig. 4. Average points banked per stop across all agents.

V. DISCUSSION

A. Reward Function Asymmetry

Iterative testing during development revealed a fundamental problem in the reward structure that drove the agent toward a conservative approach similar to the Conservative heuristic.Early reward configurations encouraged hyper aggressive drawing, producing high bust rates and poor win rates. Subsequent introduction of tiered bust penalties successfully reduced busting but created an unintended consequence as the agent learned that stopping and banking any available points was reliably rewarded, while the only signal for winning the game was a single bonus applied once per game.

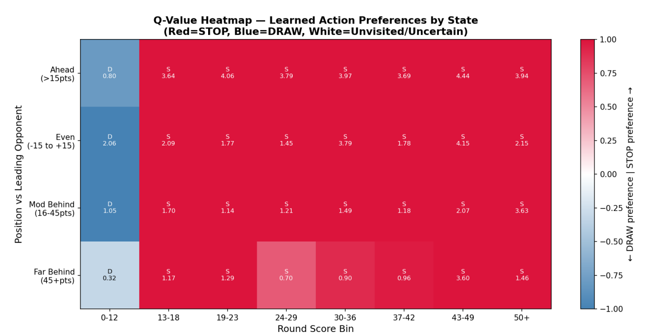

Fig. 5. Q-value heatmap showing learned action preferences by score bin and competitive position.

The win signal did not provide an incentive for winning by larger margins, winning consecutively, outperforming opponents within a round, or climbing score rankings progressively during a game. As a result the agent converged toward a conservative stopping policy that prioritized safe, incremental point accumulation over the balanced or aggressive scoring necessary to reach 200 points competitively, producing behavior weaker than the Conservative heuristic despite being trained in a competitive environment that rewards aggression.

B. State Abstraction Limitations

The original state representation attempted to track specific cards drawn by the agent to enable bust risk estimation based on deck composition. However, the resulting state space was too large for reliable Q-value estimation within a 10,000 game training budget, as the agent could not visit individual states frequently enough to build meaningful value estimates. Score binning was introduced to reduce the state space to 513 well-visited states, but this abstraction prevented the agent from reasoning about card-level risk; for example, recognizing that holding a 10 and a 12 carries higher bust exposure than holding a 2 and a 3 at the same round score. Extending the training budget to approximately 200,000 games may have allowed finer state granularity while maintaining adequate visitation frequency. Alternatively, transitioning to a Deep Q-Network architecture would allow the agent to process richer state information including specific drawn cards through neural function approximation, eliminating the dimensionality constraint inherent to tabular methods while preserving the core Q-learning framework.

C. Balanced_2 Anomaly

Initial heuristic tuning achieved win rates within 3 percentage points across all three policies in 3-player evaluation, confirming adequate calibration. However, introducing a fourth player slot filled by a second Balanced agent produced an unexpected result due to the Balanced_2 profile achieving a 35.8% win rate over 5,000 games despite running an identical policy to the Balanced agent at 23.1%. Repeated baseline runs consistently reproduced this gap, ruling out random variance as the cause.

Three hypotheses are proposed to explain this anomaly. First, the engine may not be treating both Balanced profiles as fully independent entities, creating an unintended behavior despite identical policy logic. Second, special card targeting heuristics may disproportionately attack leading players such as Greedy and Balanced, allowing Balanced_2 to accumulate points undisturbed late in games. Third, and most likely, the win condition implementation awards victory immediately upon any player reaching 200 points rather than completing the round, meaning the player acting earliest in the final round gains a first-mover advantage. Balanced_2 occupying seat zero may benefit consistently from this mechanic. Addressing these hypotheses represents a clear direction for future work, including implementing distinct heuristic policies for each agent slot and correcting the win condition to complete the full round before declaring a winner.

VI. Conclusion

This project demonstrates that a tabular Q-learning agent can learn position-sensitive stopping behavior in a stochastic multiplayer card game without explicit programming of decision thresholds. The agent achieved a win rate of 15.3%, demonstrating the capacity to outperform opponents in individual games, however it was unable to consistently outperform any of the three heuristic baselines across evaluation. The results highlight two fundamental limitations of the approach. First, the reward function structure, where immediate bust penalties outweighed few game-level win signals, drove the agent toward hyper-conservative stopping behavior that limited competitive scoring. Second, tabular state abstraction prevented card-level risk reasoning that would require either a significantly larger training budget or a transition to a more powerful reinforcement learning architecture.

Three directions are identified for future work. The highest priority is transitioning to a Deep Q-Network architecture, which would allow richer state representations including specific drawn cards without the dimensionality constraints of tabular methods. Second, the reward structure should be redesigned to provide denser winning signals, including round-level ranking rewards and progressive score gap incentives, to encourage the agent to prioritize game-winning behavior over conservative point banking. Third, the evaluation framework should be extended by introducing distinct heuristic policies for each agent slot and testing win rate sensitivity across varying player counts to eliminate positional bias and provide a more rigorous competitive benchmark.

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